

# Supplementary materials: Lexical statistics and the shape bias

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This document supplements the main manuscript. Contents: (S1) trial wording for the online rating tasks; (S2) Study 1A lexicon composition, item-count reconciliation across studies, demographics, rating distributions, and missing-identifier sensitivity; (S3) full 16-language density and Euler grids; (S4) cross-sample shape correlations; (S5) Korean age-of-acquisition models; (S6) Study 2A strict- vs. broad-shape coding; (S7) pooled English rater counts by procedure; (S8) leave-one-language-out refits of the Study 4 model; (S9) Bayesian posterior and power sensitivity for the shape–AoA coefficient.

## S1. Trial wording (Studies 1A and 2B)

Interactive browser implementations of the rating tasks accompany the Open Materials for this manuscript. The procedures match the main text.

**Study 1A.** Participants completed three blocks in random order (solidity, count–mass syntax, category-organizing feature), 20 nouns per block. Solidity and count–mass trials presented a single unilemma as the prompt. The organizing-feature trial asked whether the word *belongs to a category that is organized by* shape, color, material, or none of these (single forced choice). Words were not grouped by semantic category.

**Study 2B.** Trial structure matched Study 1A for the corresponding blocks. Instructions used the clarified organizing-feature wording described in the main text.

## S2. Study 1A: Lexicon composition, demographics, and rating diagnostics

Per-language unilemma counts and semantic categories

Item-count reconciliation across studies

Pairwise overlap of language-specific unilemma sets

Participant demographics (Study 1A)

Distributions of rating totals and response proportions

Missing participant identifiers: sensitivity analysis

A session-logging error left the participant identifier missing on 13.8% of retained Study 1A trials (main text, Study 1A Participants). All unattributed trials fall in the category-organization block (32.2% of that block); solidity and count–mass proportions are identical under either inclusion rule. Item-level mean shape

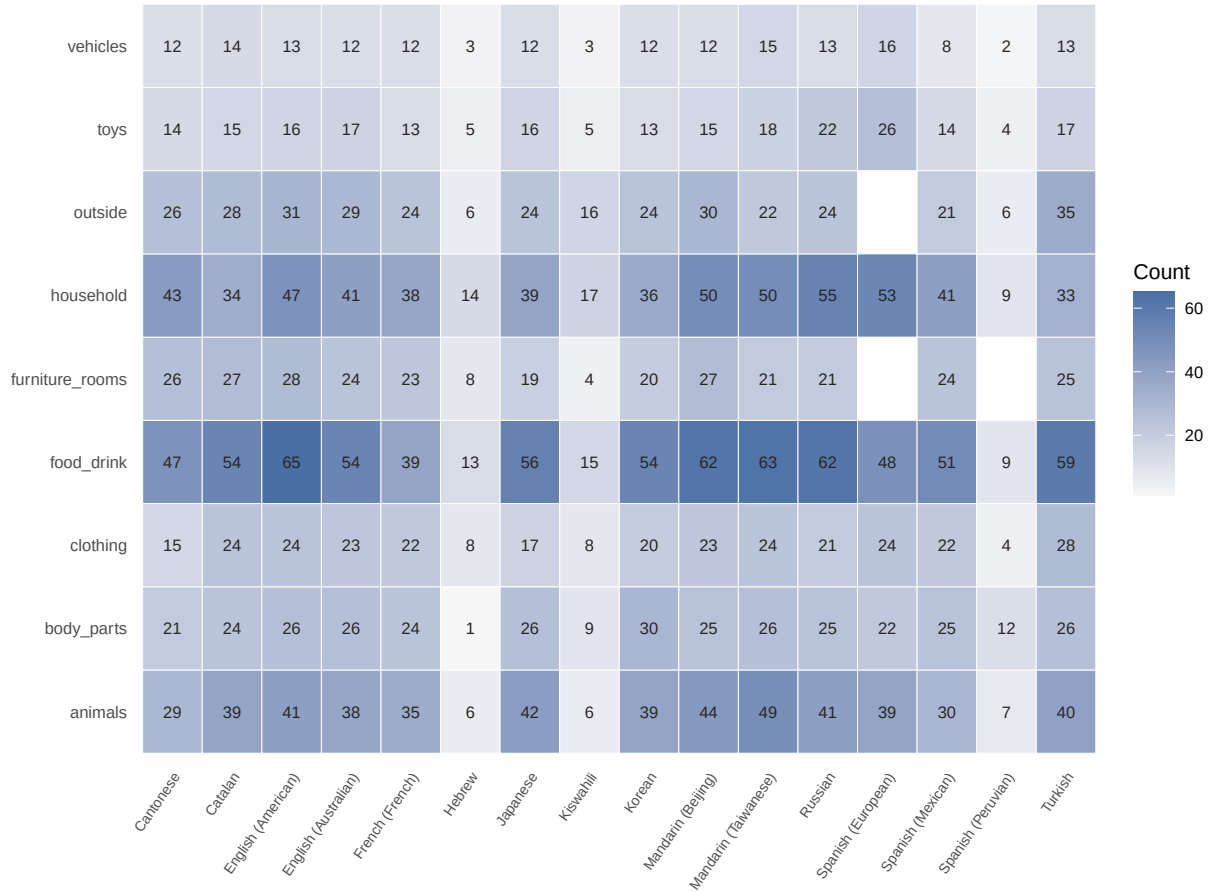


Figure 1: Semantic-category counts per language for Study 1A items (Wordbank semantic coding), for all 16 rated languages. Cell values are unlemma counts.

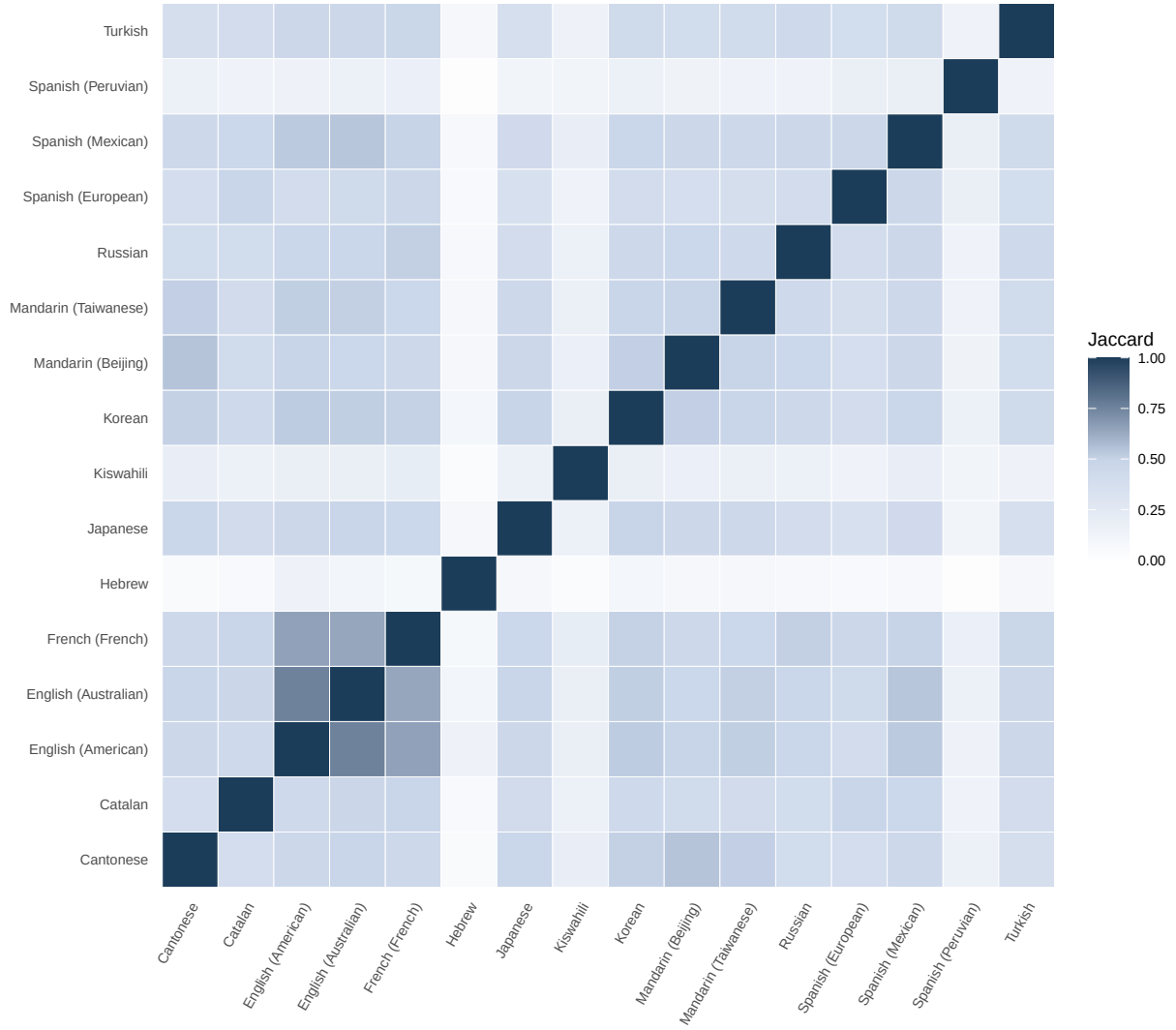


Figure 2: Pairwise Jaccard similarity between Study 1A noun lists (distinct unlemmas per language). Diagonal = 1.

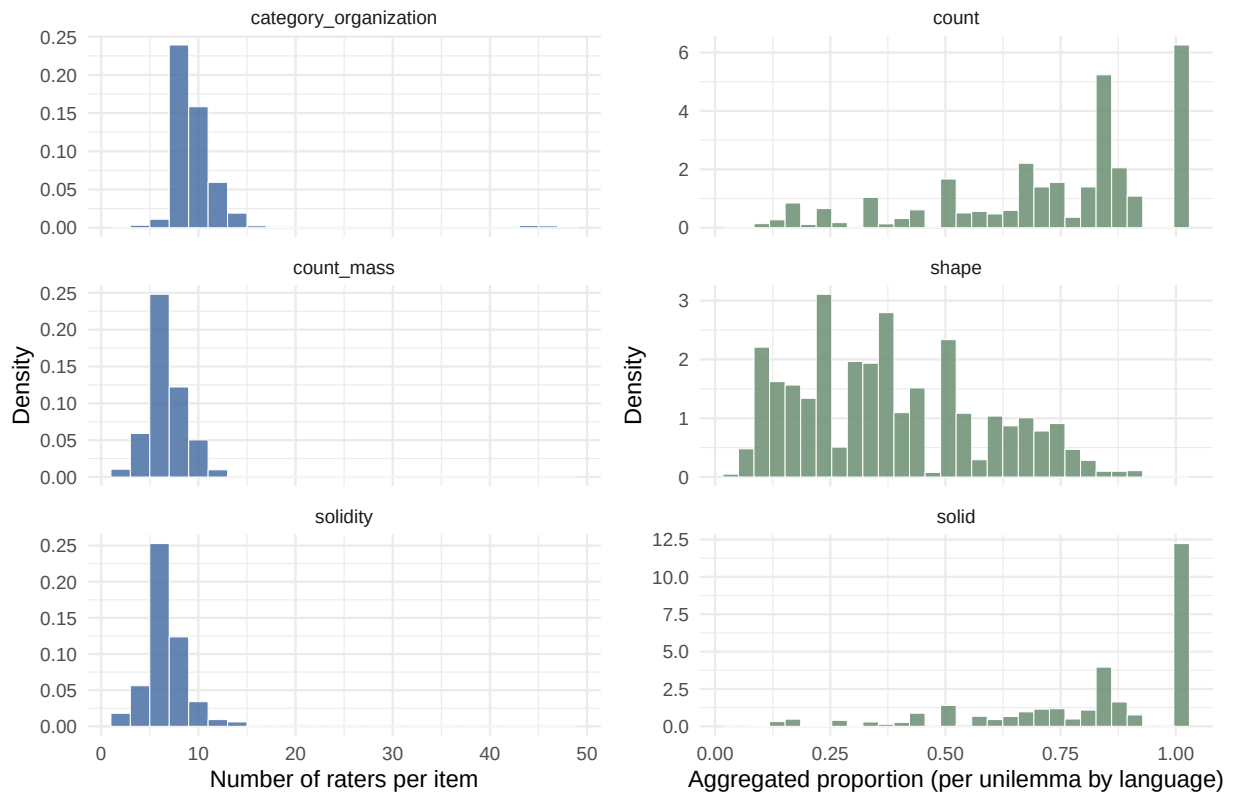


Figure 3: **Left:** Per-item rating totals by experimental block. **Right:** Aggregated item-level proportions for solidity, count-noun, and shape responses.

Table 1: Study 1A: Number of distinct unilemmas rated per Wordbank language list (English-speaking raters).

| language             | N unilemmas |
|----------------------|-------------|
| Cantonese            | 233         |
| Catalan              | 257         |
| English (American)   | 291         |
| English (Australian) | 264         |
| French (French)      | 229         |
| Hebrew               | 64          |
| Japanese             | 250         |
| Kiswahili            | 82          |
| Korean               | 247         |
| Mandarin (Beijing)   | 286         |
| Mandarin (Taiwanese) | 288         |
| Russian              | 282         |
| Spanish (European)   | 227         |
| Spanish (Mexican)    | 234         |
| Spanish (Peruvian)   | 53          |
| Turkish              | 275         |

Table 2: Item counts by sample and block: distinct items with at least one retained rating.

| Sample                        | Block              | Items |
|-------------------------------|--------------------|-------|
| Study 1A (English raters)     | Solidity           | 661   |
| Study 1A (English raters)     | Count-mass         | 663   |
| Study 1A (English raters)     | Organizing feature | 664   |
| Study 1B (English raters)     | Organizing feature | 624   |
| Study 2A (Korean, in-country) | Solidity           | 256   |
| Study 2A (Korean, in-country) | Count-mass         | 256   |
| Study 2A (Korean, in-country) | Organizing feature | 256   |
| Study 2A, unilemma-matched    | All blocks         | 242   |
| Study 2B (Korean, online)     | Solidity           | 246   |
| Study 2B (Korean, online)     | Count-mass         | 247   |
| Study 2B (Korean, online)     | Organizing feature | 246   |

*Note.* Study 1A blocks differ by a few nouns because not every noun retained ratings in every block. The Study 1B word list lacked 40 of the 664 Study 1A nouns (main text, Study 1B Materials). Study 2A items are English glosses of the Korean words (two duplicate-gloss items excluded); the unilemma-matched row counts the glosses matched to Korean CDI unilemmas that enter cross-study analyses. Study 2B counts unilemmas after the same duplicate-gloss exclusion. Pooled English counts in S7 exceed 664 because the Study 1B and Study 3 lists contain nouns absent from Study 1A.

is 0.31 with identifier-present trials only versus 0.29 with all trials (item-level  $r = 0.93$ ). Unattributed trials selected shape less often (23% vs. 32%), so missingness may relate to responses. Under either rule,

Table 3: Study 1A demographics for participants matched to the analyzed rating data. Platform exports also include returned and timed-out submissions with no analyzed trials. The main-text N (336) counts distinct recorded identifiers in the rating data; one identifier has no matching demographic record, so demographic tables sum to 335. A session-logging error additionally left the identifier missing on a subset of retained trials (main text, Study 1A Participants).

| Statistic                                    | Value |
|----------------------------------------------|-------|
| Platform submissions (all statuses)          | 639   |
| Approved submissions                         | 574   |
| Participants matched to analyzed rating data | 335   |
| Mean age (years)                             | 41.54 |
| SD age                                       | 14.29 |
| Min age                                      | 18    |
| Max age                                      | 79    |

Table 4: Study 1A: Self-reported sex (analyzed participants). One matched participant has demographic fields marked as consent-revoked in the platform export and contributes no age information.

| sex               | n   | pct   |
|-------------------|-----|-------|
| CONSENT_REVOKED   | 1   | 0.3%  |
| Female            | 202 | 60.3% |
| Male              | 131 | 39.1% |
| Prefer not to say | 1   | 0.3%  |

mean shape remains far below solidity and count-noun proportions (0.72 and 0.68). Study 1A analyses are item-aggregated; trial-level models with participant random effects appear only in Study 3 on separate samples.

Two further checks apply the identifier-present rule to the analyses downstream of the Study 1A shape measure. The primary Study 4 cross-linguistic model, refit with the identifier-present shape measure, gives  $B = -0.29$  ( $SE = 0.11$ ,  $p = .011$ ), indistinguishable from the all-trials estimate in the main text. The per-language 80%-criterion shape sets grow under the identifier-present rule (up to 14 nouns per language versus 6 with all trials) because fewer raters per item make an 80% supermajority easier to reach; they remain far smaller than the corresponding solid and count sets.

Table 5: Study 1A item-level proportions from all retained trials versus identifier-present trials only.

| Statistic  | Items  | Mean (all trials) | Mean (ID present) | Item-level r | Max  diff |
|------------|--------|-------------------|-------------------|--------------|-----------|
| Shape      | 665.00 | 0.292             | 0.311             | 0.935        | 0.417     |
| Solid      | 662.00 | 0.719             | 0.719             | 1.000        | 0.000     |
| Count noun | 664.00 | 0.678             | 0.678             | 1.000        | 0.000     |

*Note.* Missing identifiers fall only in the category-organization block, so solidity and count-noun rows are identical by construction.

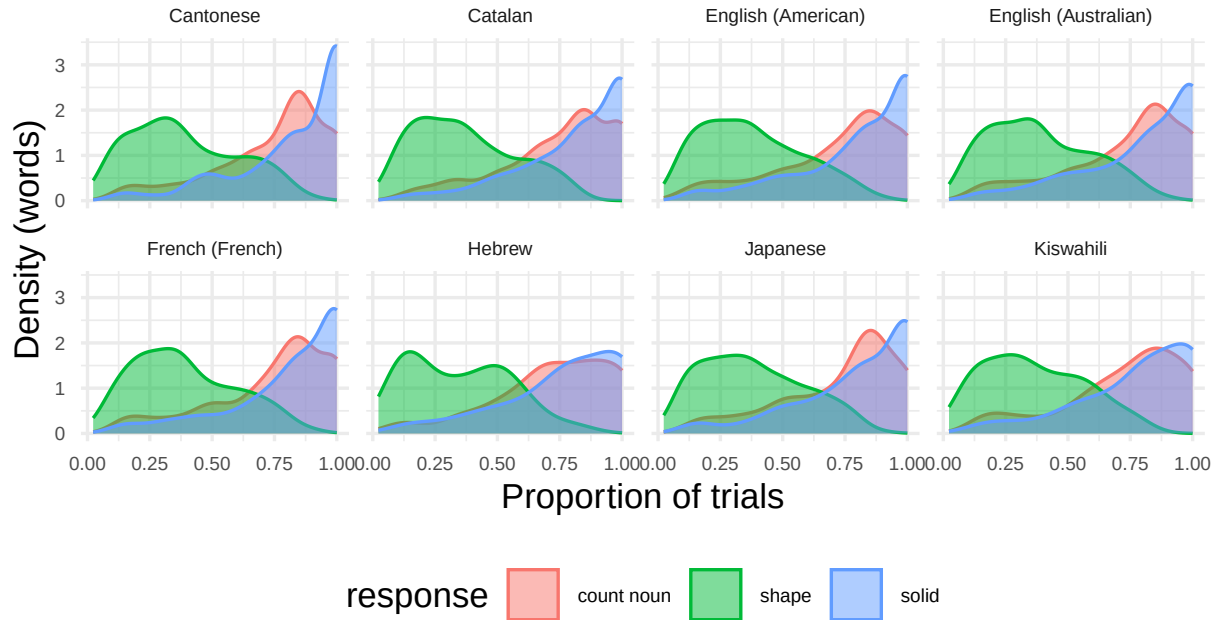


Figure 4: (Part 1 of 2) Kernel density of per-word proportions for solidity, count, and shape, Study 1A (eight languages, alphabetical).

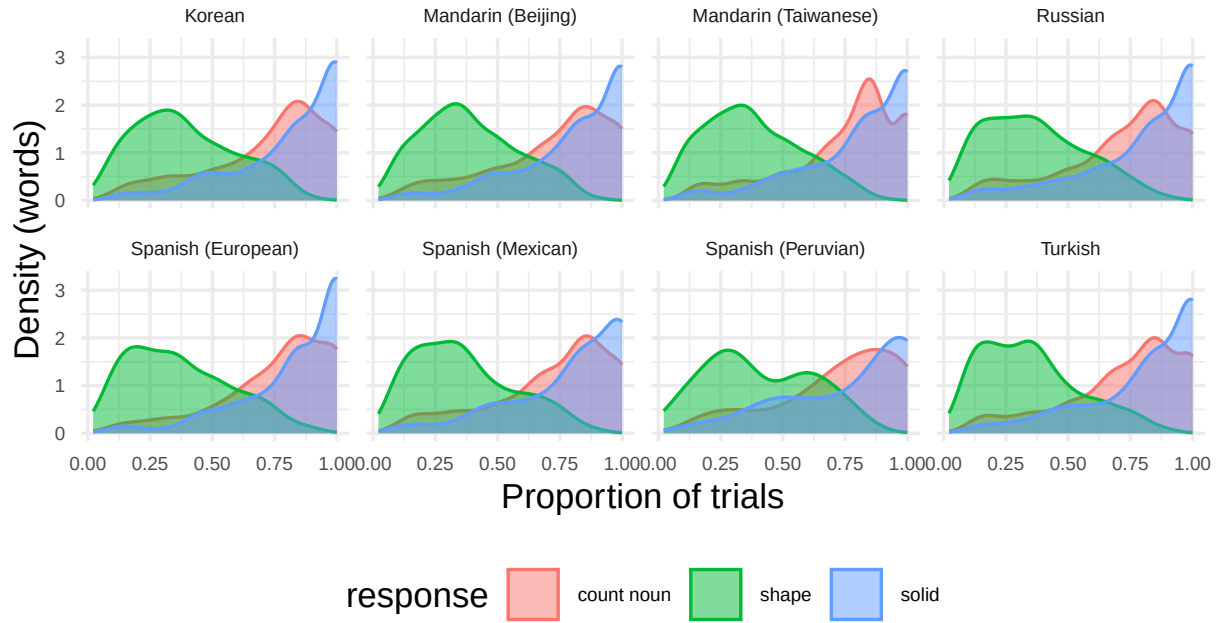


Figure 5: (Part 2 of 2) Same as Part 1 for the remaining eight languages.

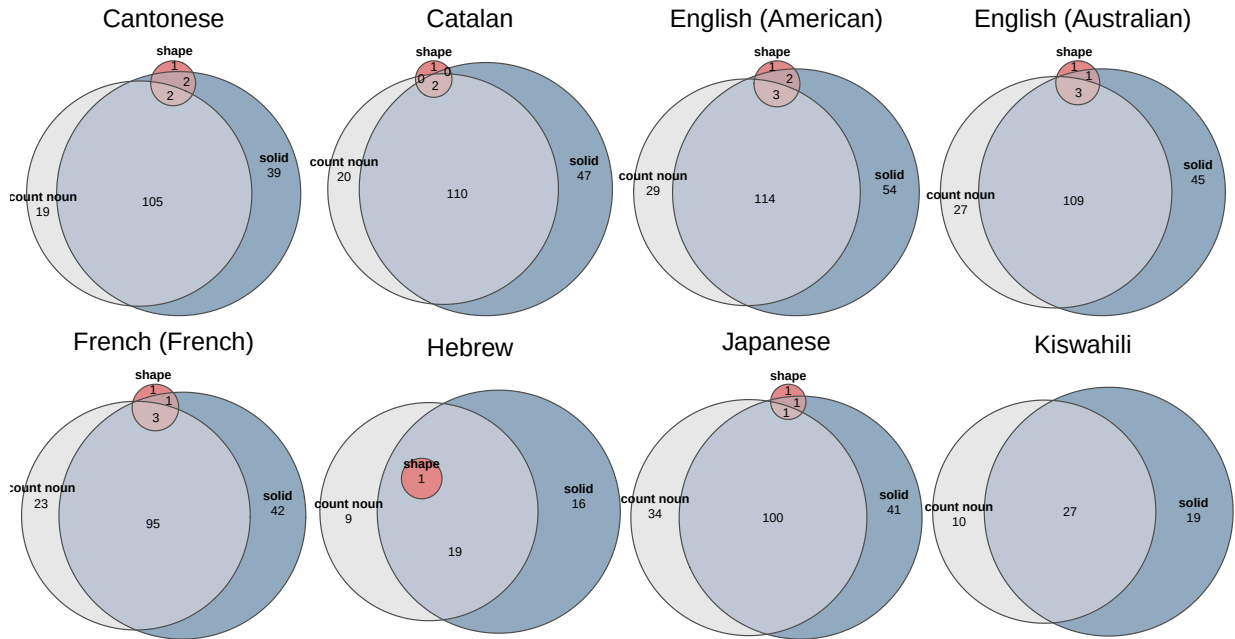


Figure 6: (Part 1 of 2) Area-proportional Euler diagrams (80% cutoff) for solid, count, and shape-organization, Study 1A (eight languages; same order as the density figure).

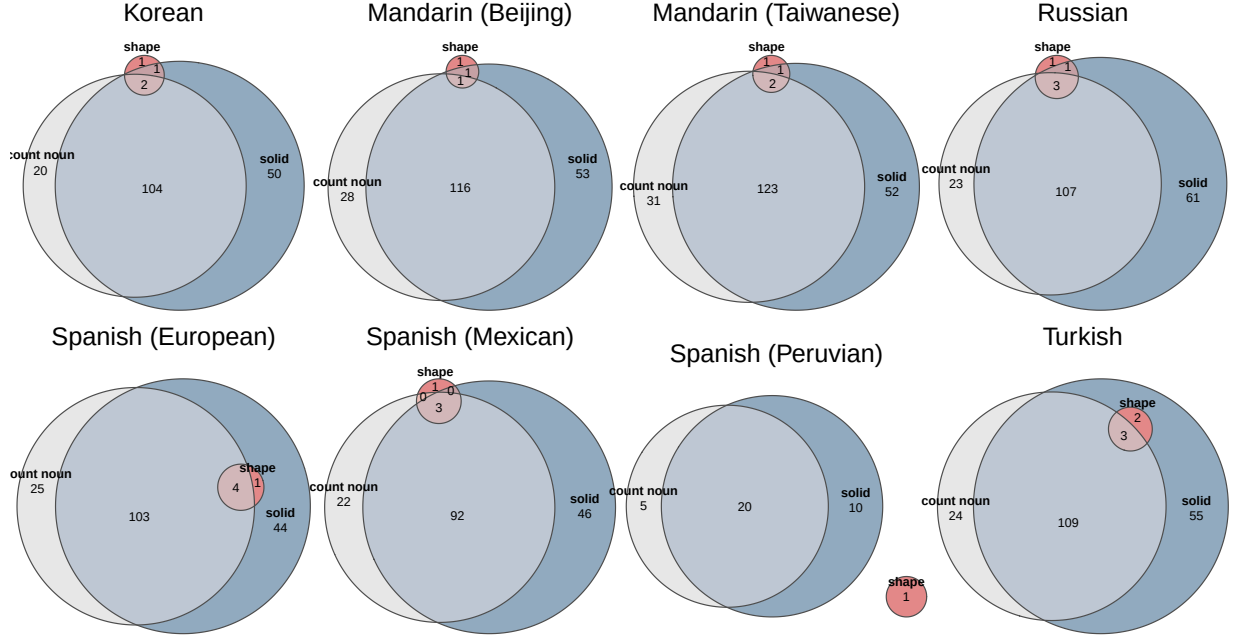


Figure 7: (Part 2 of 2) Area-proportional Euler diagrams for the remaining eight languages.

Table 6: Pearson correlations of word-level shape-organization proportions across the four norming samples (Study 2A = in-country Korean; Study 2B = online Korean).

| Sample   | Study 1A | Study 1B | Study 2B | Study 2A |
|----------|----------|----------|----------|----------|
| Study 1A | 1.00     | 0.65     | 0.47     | 0.49     |
| Study 1B | 0.65     | 1.00     | 0.48     | 0.46     |
| Study 2B | 0.47     | 0.48     | 1.00     | 0.51     |
| Study 2A | 0.49     | 0.46     | 0.51     | 1.00     |

### S3. Study 1A: Sixteen-language grids

### S4. Cross-sample shape correlations

### S5. Korean age-of-acquisition models

On the matched Korean Wordbank AoA item set, two models share the same rows and differ in which lexical-feature ratings enter. The English-predictor model includes Korean corpus log frequency, Brysbaert concreteness, and Study 1A/1B solidity, shape, and count-noun ratings. The Korean-predictor model includes the same Korean corpus log frequency and Korean-averaged solidity, shape, and count-noun ratings (Studies 2A and 2B), omitting concreteness because those norms are English-lemma ratings. Only a category random intercept is identifiable.

Because concreteness enters only the English-predictor model (the norms are English ratings), we also refit that model without it so the two models share the same covariate set; the comparison is unchanged (AIC 844.2 vs. 835.4 for the Korean-predictor model).

Table 7: Fixed effects for the English-predictor model of Korean age of acquisition (months).

| Predictor    | Estimate | SE   | df    | t      | p      |
|--------------|----------|------|-------|--------|--------|
| (Intercept)  | 25.86    | 0.23 | 4.3   | 112.04 | < .001 |
| logfreq      | -2.11    | 0.20 | 126.9 | -10.36 | < .001 |
| concreteness | -0.03    | 0.20 | 164.4 | -0.14  | .891   |
| solid        | 0.21     | 0.23 | 172.5 | 0.89   | .373   |
| mean_shape   | -0.24    | 0.22 | 75.8  | -1.12  | .266   |
| count_noun   | -0.08    | 0.25 | 162.5 | -0.33  | .742   |

*Note.* Random intercept for semantic category. Predictors are z-scored. N = 175. AIC = 846.1; marginal  $R^2 = 0.40$ .

Table 8: Fixed effects for the Korean-predictor model of Korean age of acquisition (months).

| Predictor      | Estimate | SE   | df    | t      | p      |
|----------------|----------|------|-------|--------|--------|
| (Intercept)    | 25.81    | 0.27 | 7.5   | 95.50  | < .001 |
| logfreq        | -2.13    | 0.20 | 160.8 | -10.66 | < .001 |
| mean_solid_kor | 0.56     | 0.25 | 169.5 | 2.24   | .026   |
| mean_shape_kor | 0.27     | 0.23 | 93.9  | 1.15   | .255   |
| mean_count_kor | -0.83    | 0.25 | 174.8 | -3.26  | .001   |

*Note.* Random intercept for semantic category. Predictors are z-scored. N = 175. AIC = 835.4; marginal  $R^2 = 0.45$ . English-predictor model on the same items: AIC 846.1; marginal  $R^2 = 0.40$ .

## S6. Study 2A: Strict vs. broad shape coding

Strict coding counts shape only when it was the sole selected feature; broad coding counts shape whenever it was selected alone or with another feature.

Table 9: Study 2A multiple-selection coding. Values match the main text.  $P(\text{shape})$  = item-level shape proportion under each coding rule.

| Description                          | Value |
|--------------------------------------|-------|
| Organizing-block trials (weighted N) | 7,004 |
| Shape only (strict)                  | 4,205 |
| Shape + other feature (broad only)   | 2,093 |
| N items                              | 242   |
| Mean $P(\text{shape})$ , strict      | 0.587 |
| SD, strict                           | 0.159 |
| Mean $P(\text{shape})$ , broad       | 0.899 |
| SD, broad                            | 0.169 |

## S7. Pooled English rater counts

Organizing-feature ratings were pooled across English samples that shared a procedure, then re-scored under the 80% rule. Solidity and count-mass criteria remain from Study 1A (the only sample that collected those blocks).

Pooling Studies 1A, 1B, and the forced-choice random cell of Study 3 raised the median number of organizing-feature raters per noun from 9 to 22, left mean shape slightly lower, and changed 80% shape counts only marginally. Pooled word counts exceed the 664-noun Study 1A set because the Study 1B and Study 3 lists contain nouns absent from Study 1A; pooled rows count every word with at least one rating in any contributing sample.

Table 10: Organizing-feature trials by procedure. FC = forced choice with random word order; MS blocked = multiple selection with semantic grouping.

| Procedure                        | Trials | Words  | Median raters/word | Mean shape |
|----------------------------------|--------|--------|--------------------|------------|
| Study 1A (FC random)             | 6,856  | 665.00 | 9.00               | 0.29       |
| Pooled FC random (1A + 1B + 2x2) | 16,129 | 749.00 | 22.00              | 0.27       |
| MS blocked pilot                 | 1,399  | 659.00 | 2.00               | 0.57       |
| Pooled MS blocked (pilot + 2x2)  | 3,759  | 742.00 | 5.00               | 0.57       |

Table 11: 80% cutoff counts in four languages: Study 1A shape ratings only versus pooled forced-choice random-order shape ratings. Solid and count criteria are from Study 1A in both rows.

| Pool             | Language           | N shape | N intersect | Median raters | Mean shape |
|------------------|--------------------|---------|-------------|---------------|------------|
| Study 1A only    | English (American) | 5.00    | 3.00        | 9.00          | 0.34       |
| Study 1A only    | Japanese           | 3.00    | 1.00        | 10.00         | 0.35       |
| Study 1A only    | Korean             | 3.00    | 2.00        | 10.00         | 0.34       |
| Study 1A only    | Mandarin (Beijing) | 2.00    | 1.00        | 9.00          | 0.35       |
| Pooled FC random | English (American) | 5.00    | 3.00        | 23.00         | 0.32       |
| Pooled FC random | Japanese           | 4.00    | 2.00        | 23.00         | 0.33       |
| Pooled FC random | Korean             | 6.00    | 4.00        | 23.00         | 0.33       |
| Pooled FC random | Mandarin (Beijing) | 5.00    | 3.00        | 23.00         | 0.33       |

On the pooled forced-choice trials,  $n \in \{7, 9, 11, 13\}$  raters were redrawn per noun. Even at  $n = 11$ , fewer than 3% of items passed the 80% shape rule on average.

Table 12: Subsample sensitivity on pooled forced-choice random-order trials. Mean pass rate = share of 400 resamples per word reaching  $\geq 80\%$  shape; usually passing = pass rate  $\geq .5$ .

| Raters per word | Eligible words | Mean pass rate | Words usually passing |
|-----------------|----------------|----------------|-----------------------|
| 7.00            | 722.00         | 2.8%           | 8.00                  |
| 9.00            | 701.00         | 1.5%           | 4.00                  |
| 11.00           | 679.00         | 2.2%           | 8.00                  |
| 13.00           | 651.00         | 1.3%           | 4.00                  |

Pooling the grouped multiple-selection pilot and the Study 3 grouped cell yields mean broad shape near the Study 3 cell estimates, but only 29.3% of Study 1A nouns meet an 80% broad-shape cutoff, because most items in the  $2 \times 2$  cells have few raters each.

Table 13: Multiple-selection grouped samples (broad coding). The on-1A columns restrict to nouns that appear in Study 1A.

| Pool              | Words  | Words on 1A set | Pct shape $\geq 80$ | Pct on 1A set | Median raters | Mean broad shape |
|-------------------|--------|-----------------|---------------------|---------------|---------------|------------------|
| MS grouped pilot  | 659.00 | 583.00          | 39.90               | 41.90         | 2.00          | 0.57             |
| Pooled MS grouped | 742.00 | 659.00          | 28.30               | 29.30         | 5.00          | 0.57             |

## S8. Leave-one-language-out refits of the Study 4 model

The primary cross-linguistic model of Study 4 was refit 14 times, each time dropping all items contributed by one language. The shape coefficient stayed negative in every refit and varied between -0.31 and -0.26, against a pooled estimate of -0.28, and the Wald test fell below .05 in 14 of the 14 refits. No single language’s item set is necessary for the association, although this check retains nearly all unilemmas on each pass (the shape predictor is unilemma-level), so it speaks to language-specific item sets rather than to the unilemma-level predictor; the more stringent leave-one-category-out refits appear in the main text (Study 4 Results).

Table 14: Leave-one-language-out refits of the Study 4 cross-linguistic model. Each row drops one language’s item-by-language observations and reports the shape coefficient from the refit (months of age of acquisition per SD of shape rating).

| Language dropped     | N obs | B (shape) | SE   | 95% CI         | p    |
|----------------------|-------|-----------|------|----------------|------|
| none (pooled model)  | 3,420 | -0.28     | 0.11 | [-0.50, -0.06] | .013 |
| Cantonese            | 3,188 | -0.31     | 0.11 | [-0.53, -0.09] | .006 |
| Catalan              | 3,164 | -0.27     | 0.11 | [-0.50, -0.05] | .016 |
| English (American)   | 3,129 | -0.29     | 0.11 | [-0.51, -0.06] | .012 |
| English (Australian) | 3,156 | -0.28     | 0.11 | [-0.50, -0.06] | .013 |
| French (French)      | 3,191 | -0.29     | 0.11 | [-0.51, -0.07] | .011 |
| Hebrew               | 3,356 | -0.27     | 0.11 | [-0.49, -0.05] | .016 |
| Japanese             | 3,170 | -0.28     | 0.11 | [-0.50, -0.06] | .014 |
| Korean               | 3,173 | -0.30     | 0.11 | [-0.52, -0.08] | .008 |
| Mandarin (Beijing)   | 3,137 | -0.28     | 0.12 | [-0.51, -0.05] | .016 |
| Mandarin (Taiwanese) | 3,133 | -0.27     | 0.11 | [-0.49, -0.06] | .012 |
| Russian              | 3,138 | -0.26     | 0.11 | [-0.48, -0.03] | .026 |
| Spanish (European)   | 3,193 | -0.29     | 0.11 | [-0.52, -0.07] | .010 |
| Spanish (Mexican)    | 3,187 | -0.28     | 0.11 | [-0.51, -0.06] | .013 |
| Turkish              | 3,145 | -0.27     | 0.11 | [-0.49, -0.05] | .016 |

*Note.* First row is the pooled model reported in the main text. Wald 95% confidence intervals. All other model terms, including the language fixed effect and the crossed category and unilemma random intercepts, are as in the main model.

## S9. Bayesian refit and power sensitivity for the shape–AoA coefficient

Both models were refit in `brms` with weakly informative normal(0, 5) priors on the regression coefficients, four chains of 4,000 iterations (1,000 warmup), and `adapt_delta` = 0.95. The region of practical equivalence

(ROPE) is  $\pm 0.1$  SD of age of acquisition, or  $\pm 0.33$  months. For the primary cross-linguistic model, 99.3% of the posterior mass for the shape coefficient lies below zero and 32.9% lies below the ROPE. The direction of the effect is therefore well supported while its practical size is not: about 67% of the posterior is compatible with a negligible effect.

Table 15: Posterior summaries for the shape coefficient in the Bayesian refits of the Study 4 models.

| Model                      | Post. mean | Post. SD | 95% CrI        | $P(B < 0)$ | % in ROPE | % below ROPE | Max $\hat{R}$ |
|----------------------------|------------|----------|----------------|------------|-----------|--------------|---------------|
| cross-linguistic (primary) | -0.28      | 0.11     | [-0.50, -0.06] | 0.993      | 67.1      | 32.9         | 1.001         |
| English (American) only    | -0.40      | 0.17     | [-0.74, -0.06] | 0.990      | 33.6      | 66.4         | 1.000         |

*Note.* Coefficients are in months of age of acquisition per SD of shape rating. CrI = credible interval; ROPE = region of practical equivalence,  $\pm 0.1$  SD of age of acquisition.

The single-language design cannot arbitrate the same question. Simulating age of acquisition from the English-only design ( $N = 291$ ) at a range of true slopes, with the fitted category-intercept and residual variances, gives the rejection rates below.

Table 16: Simulated power of the English-only Study 4 design ( $N = 291$  items, 500 simulations per row) to detect a shape effect of a given size.

| True shape slope | Power |
|------------------|-------|
| -0.28            | 0.42  |
| -0.41            | 0.66  |
| -0.50            | 0.84  |
| -0.75            | 0.99  |
| -1.00            | 1.00  |
| -1.50            | 1.00  |

*Note.* Slopes are in months per SD of shape rating; the pooled cross-linguistic estimate is -0.28. The analytic minimal detectable effect at 80% power is 0.48 for the English-only design and 0.67 for the Korean-predictor design.